

Week 4 Day 4: Tuning Report

Model Tuning, Regularization & Reproducible Pipelines

Hyperparameter Search

The two best models from the previous day, **Logistic Regression** and **LightGBM**, were tuned using `RandomizedSearchCV` with 5-fold Stratified Cross-Validation. Since my primary evaluation metric is **Precision**, the search was optimized using Precision instead of Accuracy or F1.

For Logistic Regression, the best parameters were L1 regularization with $C = 0.001$. This showed that a simpler model with stronger regularization performed better than a more complex one.

For LightGBM, the best parameters were `learning_rate = 0.01`, `max_depth = 3`, `n_estimators = 100`, and `l2_regularization = 1.0`. These settings produced the highest cross-validated Precision while keeping the model well controlled and reducing overfitting.

	Model	Best Precision
0	Logistic Regression	0.769856
1	LightGBM	0.981490

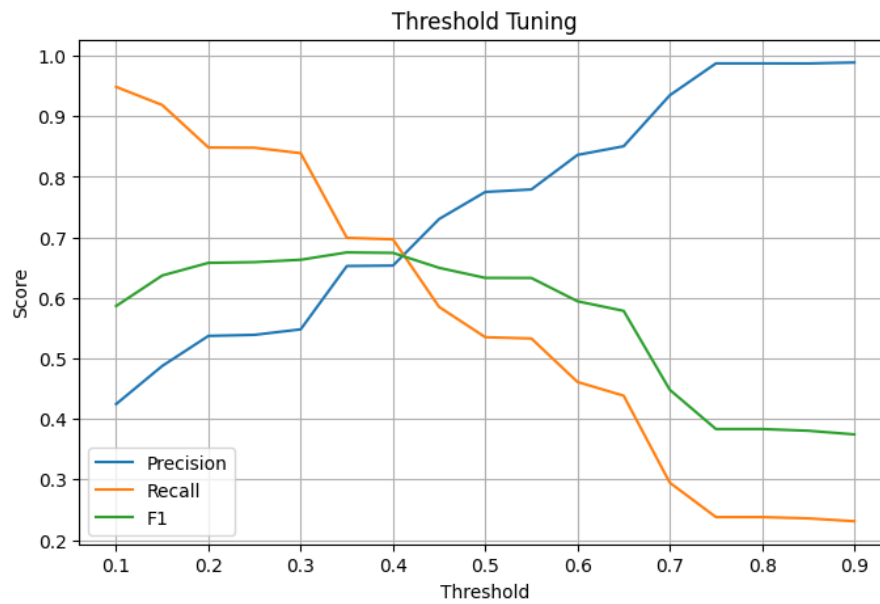
Model Diagnostics

Learning curves were used to check whether the models were overfitting or underfitting. Both **Logistic Regression** and **LightGBM** showed training and validation curves that stayed close together and leveled off as more training data was added. This indicates **low variance**, meaning the models generalize well without overfitting. Since both models also achieved consistently good Precision, they also show **low bias** and are able to learn the important patterns in the data.

The effect of **C** on Logistic Regression showed that stronger regularization gave the best validation performance, while larger values of **C** slightly reduced generalization. For **LightGBM**, increasing **max_depth** beyond 5 improved the training score but reduced validation Precision, showing the start of overfitting. Keeping the trees shallow was therefore the better choice.

Probability calibration was then evaluated using calibration plots and the **Brier Score**. The original model achieved a **Brier Score of 0.1191**, showing that its predicted probabilities were already fairly reliable. After comparing **Sigmoid** and **Isotonic** calibration, **Isotonic** followed the ideal calibration line more closely, so it was chosen as the final calibration method.

Different classification thresholds were also tested. Lower thresholds gave higher Recall but more false positives, while higher thresholds increased Precision but missed many high-income customers. I chose a threshold of **0.60** because it provided the best balance. At this threshold, the model achieved **83.63% Precision** while still identifying **46.11%** of the high-income individuals. Although higher thresholds gave even better Precision, they missed too many potential customers to be practical.



Final Model Performance

The final model is a tuned and isotonic-calibrated LightGBM pipeline using a classification threshold of 0.60. It achieved the following results on the untouched hold-out test set:

Metric	Score
Accuracy	0.8485
Precision	0.8363
Recall	0.4611
F1 Score	0.5944
ROC AUC	0.8942

The confusion matrix showed that the model correctly identified 1,078 high-income individuals while producing 211 false positives. Although this is more false positives than higher thresholds, it also captures almost twice as many high-income customers compared to a threshold of 0.75. This makes 0.60 a much better balance between Precision and Recall while still keeping false positives reasonably low.

Conclusion

Overall, tuning helped improve the final model and made its predictions more reliable.

LightGBM remained the best-performing model after tuning, and **Isotonic calibration** made its probability estimates more accurate. Choosing a threshold of **0.60** gave a better balance between Precision and Recall, making the model more useful in practice while still keeping false positives low. The final pipeline is reproducible, saved, and ready to use on new data.